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HW 1

1

Note

All of these exercises come from Utts and Heckard [1].

HW 1

Remark. These solutions have roughly the level of writing that I expect to see in your homework. Notice that I have used complete and concise sentences to communicate not only the answers, but a reminder of what was asked. In cases where it's warranted, I give explanations for my answers. These explanations are just as important as the answers themselves.

2.3

- a) An instructor surveying all the students in her class to determine their preference is gathering data from the **entire population** that she wishes to study.
- b) A poll of 1000 people that is used to estimate a nationwide opinion on POTUS's job performance represents **sample data**, since in this case the population consists of all the nation's people (a set far exceeding 1000 in size).

2.5

- a) In a census, every member of the population is counted. Therefore, the 2000 census's estimate that the average US household size was 2.59 persons represents a **population parameter**.
- b) An ABC News poll finds that 36% of a 500-person sample of Americans supports a particular view. Since the goal is to estimate the percentage of Americans (the "population") in support of that view, the 36% is a **sample statistic**.
- c) "To estimate average normal body temperature of all adults, a doctor measures the temperatures of 100 healthy adults. The average temperature for that group is 98.2 degrees Fahrenheit." ([1, Page 55]). Here, 98.2 is a **sample statistic**, since it is calculated from sample data, and is used to estimate a population parameter.

2.11

- a) Importance of religion to respondent (very, somewhat, or not very important): **Categorical**. The categories are the three listed in parentheses.

- b) Hours of sleep last night. **Quantitative**. One must measure this numerical quantity with a timer.
- c) Weights of adult women, measured in pounds. **Quantitative**. Again, it's a numerical quantity which must be measured.
- d) Favorite color for an automobile. **Categorical**. The categories are the various colors.

2.21

A physiologist records the pulse rates of 30 men and 30 women.

- a) The two variables here are **sex** and **pulse rate**.
- b) Sex is a **categorical** variable, its values being non-numerical labels. Pulse rate is a **quantitative** variable, it being a number which must be measured.
- c) In order to compare pulse rates of men and women, one might ask "Do men and women have the same pulse rates on average?" It would then be useful to compute the average pulse rate $\bar{x}$ found for the 30 men, and the average pulse rate $\bar{y}$ found for the 30 women. One could then compare $\bar{x}$ and $\bar{y}$.

2.22

Most of the questions in the surveys students fill out at the end of a semester are ordinal. For example, *Please rate the overall quality of the course: Poor, Fair, Good, Outstanding, Exceptional*. Here, there are five categories, and they have a natural ranking from worst to best. But it would not be meaningful to average the results of the students' answers.

2.25

- a) In order to answer the question *Is the average IQ of left-handed people higher than the average IQ of right-handed people?*, researchers should gather data for the variables of **IQ** and **handedness**. IQ is a quantitative variable whose values are a numerical score on an IQ test. Handedness is a categorical variable whose values are "right-handed" and "left-handed." (The question, as posed, disregards the existence of the category "ambidextrous.") The question views handedness as the explanatory variable, and IQ as the response variable.
- b) In the question *For married couples, is there a relationship between owning a pet and whether or not they get divorced?*, the two variables of interest are the categorical variables of **pet ownership** (with categories pet-owner and non-pet-owner) and **divorce** (with categories they-get-divorced and they-stay-together). Pet ownership is the explanatory variable, and divorce is the response variable.

2.30

With reference to a study comparing the grades of 12th grade students and their responses to the question of how often they wear a seatbelt (see the table on [1, Page 57]):

- a) Of the 2470 students in the table

$$\frac{1700}{2470} \times 100 \approx 68.8\%$$

said that they usually get A's and B's.

- b) Of the 1700 students who said that they usually get A's and B's,

$$\frac{1056}{1700} \times 100 \approx 62.1\%$$

said that they always wear a seatbelt.

- c) Of the 657 students who said that they usually get C's,

$$\frac{300}{657} \times 100 \approx 45.7\%$$

said that they always wear a seatbelt.

- d) Of the 113 students who said that they usually get D's and F's,

$$\frac{41}{113} \times 100 \approx 36.3\%$$

said that they always wear a seatbelt.

2.35

Of 92 students who responded to "Randomly pick a letter—S or Q," 61 picked S and 31 picked Q. The order of the letter choices was reversed for another 98 students who responded to "Randomly pick a letter—Q or S." Of these 98 students, 45 picked S and 53 picked Q.

- a) The following two-way table contains the data and summarizes the relationship between the letter listed first and the letter picked.

	S	Q	Total
"...S or Q."	61	31	92
"...Q or S."	45	53	98
Total	106	84	190

- b) Of the 92 students who saw S listed first, **66.3%** ($= 61/92$) picked S and the remaining **33.7%** picked Q.

- c) Of the 98 students who saw Q listed first, 45.9% ($= 45/98$) picked S and the remaining 54.1% picked Q.
- d) bar chart omitted
- e) The data suggest that the order in which the letters are listed does influence the letter that is picked.

...But how strong is the suggestion? Of the 190 students, 114, or 60%, picked the first letter. If the students were truly picking randomly (with no external influence, neither from the listing order nor anything else), we would expect something close to a 50-50 split. Is 60% lopsided enough to say that the listing order had an effect? Or does it still count as being “close to a 50-50 split”? And what to make of the fact that the bias toward the first letter was more pronounced when it was S that was listed first?

These questions are not easy to answer intuitively. We will study methods for quantifying our confidence in our answers to them. In the meantime, think about how your intuition would change if similar proportions came from sample sizes that were either much smaller or much larger. Consider the hypothetical tables

	S	Q	Total
“...S or Q.”	6	3	9
“...Q or S.”	5	5	10
Total	11	8	19

and

	S	Q	Total
“...S or Q.”	61000000	31000000	92000000
“...Q or S.”	45000000	53000000	98000000
Total	106000000	84000000	190000000

The first one would convince me of nothing, while the second one would have me believing not only that there indeed existed a first-letter bias, but also an S-bias.

2.59

We are asked for means and medians of lists of values.

- a) {64, 68, 72, 76, 80, 86}

$$\text{mean} = \frac{64 + 68 + 72 + 76 + 80 + 86}{6} = \frac{446}{6} \approx 74.3$$

$$\text{median} = \frac{72 + 76}{2} = 74$$

b) {10, 6, 2, 7, 100}

$$\text{mean} = \frac{10 + 6 + 2 + 7 + 100}{5} = \frac{125}{5} = 25$$
$$\text{median} = 7$$

c) {30, 10, 40, 30}

$$\text{mean} = \frac{30 + 10 + 40 + 30}{4} = \frac{110}{4} = 27.5$$
$$\text{median} = 30$$

2.74

We are given a list of point differences in a series of 11 football games:

$$+38, -14, +24, -13, -9, -7, -2, -11, -7, +4, +24$$

- a) The value of the mean point difference and the value of the median point difference for the 11 games are **2.45** and **-7**, respectively.
- b) Of those two summary values, **the median**, -7, does a better job of describing the season, since (I think) what really matters is whether a team wins or loses—not necessarily the size of the margin of the loss/victory. The fact that the median is a negative number here reflects that the team lost more games than it won. By contrast, the mean, 2.45, may be telling us that when the team won, it tended to win by a lot, and when it lost, it tended to lose by a smaller margin. Perhaps that serves as some consolation to the players after a losing season.

References

- [1] Utts and Heckard. *Mind on Statistics*. 5th Edition.